Devadharshini Ayyappan

PhD Student, Electrical Engineering · 3D Computer Vision

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RESEARCH FOCUS

Physically grounded 3D/4D reconstruction of deformable and articulated objects, combining **Gaussian splatting**, **differentiable physics simulation**, and **multi-view sensing systems** for real-world and synthetic data.

Interests: 3D/4D Gaussian Splatting · Neural Rendering · Inverse Physics · Physics-Based Simulation & Synthetic Data · Sensor Fusion

EDUCATION

North Carolina State University, Raleigh, NC — **GPA: 4.0/4.0**

Ph.D., Electrical Engineering (Computer Vision & Machine Learning) — *Aug 2024 – Present*

M.S., Electrical Engineering (Computer Vision & Machine Learning; transferred to Ph.D.) — *Aug 2022 – Aug 2024*

Anna University, Chennai, India — **CGPA: 8.94/10**

B.E., Electronics and Communication Engineering — *Aug 2015 – May 2019*

RESEARCH EXPERIENCE

Dolby Laboratories, Atlanta, GA — **May 2026 – Aug 2026**

Research Intern, 3D Reconstruction

- Developed **Non-Rigid Multi-Part Triplane-Based Gaussian Splatting**, extending multi-part TVGS from rigid articulation to non-rigid object deformation.
- Designed a **soft boundary blending** mechanism and a **learned residual deformation module** to model local bending, stretching, and smooth surface deformation beyond rigid joint transforms.
- Built a **Blender synthetic dataset** of deformable articulated objects (e.g., foldable cushion chairs) with multiple deformation states for training and evaluation.
- Improved reconstruction and deformation fidelity over rigid multi-part Gaussian Splatting baselines on the synthetic benchmark.

ARoS Lab, North Carolina State University, Raleigh, NC — **Apr 2023 – Present**

Graduate Research Assistant (Advisor: Prof. Edgar Lobaton)

Physical Parameter Learning & Visual Dynamics

- Learning physical and material parameters of deformable objects from visual cues using differentiable simulation (**NVIDIA Warp**, **Newton**).
- Converted Gaussian and point-based representations into **MPM** particles and meshes to simulate deformable dynamics; used synthetic reference videos of **object–force interactions** to optimize physical consistency.

3D Reconstruction & Mesh Generation

- Built **NeRF** and **Gaussian Splatting** pipelines on custom datasets and extracted high-fidelity textured meshes (SuGaR, PGSR, **UV texture mapping**).
- Fine-tuned **feed-forward** reconstruction models (VGGT) for object- and scene-level reconstruction, experimenting with memory and register tokens for efficiency.
- Built **SfM** pipelines (COLMAP) and **Isaac Sim** synthetic data generation with physics-based rendering to improve sim-to-real generalization.

Sensing Systems for Mussel Behavior Monitoring

- Designed **Raspberry Pi / ROS2** multi-view capture systems (underwater imaging, rotating-arm, and rotating-platform rigs) with **camera calibration** and cloud synchronization [P2].
- Developed a **dual-IMU** gaping-angle estimator using Rodrigues-rotation-based differential estimation, achieving **~1.9° RMSE** on real experimental data.

INDUSTRY EXPERIENCE

Mithril Defense Inc., Remote — **May 2025 – Aug 2025**

3D Reconstruction Intern

- Performed **LiDAR–fisheye camera calibration** for large-scale 3D reconstruction and high-fidelity mesh generation.
- Optimized Gaussian Splatting and mesh rendering plugins in **Unreal Engine**, reducing large Gaussian PLY file sizes for faster visualization.
- Reconstructed meshes from LiDAR point clouds using **Marching Cubes** to improve geometric accuracy and surface quality.

Robotics88, Remote **Feb 2024 – May 2024**
Computer Vision Engineer Co-op

- Implemented real-time **camera–LiDAR fusion** (Livox Mid-360, ROS Noetic) for drone operation.
- Benchmarked people detection with **YOLOv8** and **YOLO-World** (zero-shot) on recorded ROS bag data.

USDA ARS Western Regional Research Center, Albany, CA **May 2023 – Jul 2023**
Machine Learning Intern

- Converted wheat pan-genome files into a vector database and built a **LangChain + LLM** Streamlit chat interface for querying pan-genomes [P1].

Robert Bosch, Remote **Sep 2019 – Jul 2022**
Software Engineer

- Engineered automated software analyzers that improved efficiency by **30%**; managed powertrain embedded software for 12V gasoline systems under SDLC and CI/CD practices.
- Integrated software modules and ran system tests on Daimler ECUs using HIL LABCARs.

PUBLICATIONS

[P1] G. R. Lazo, **D. Ayyappan**, P. K. Sharma, V. K. Tiwari. "Contextual Science and Genome Analysis for Air-Gapped AI Research." *bioRxiv*, 2025. doi:10.1101/2025.03.21.644606

[P2] M. Snap, A. Badalamenti, **D. Ayyappan**, S. Banerjee, D. M. Urieta, C. Cooper, M. Daniele, J. Reynolds, J. Levine, A. Bozkurt, E. Lobaton. "AI-Driven Mussel Behavior Monitoring Using an Accessible 3D Imaging System." *ASEE Annual Conference & Exposition*, 2025. [NSF PAR]

[P3] J. L. Jimenez, P. Gandhi, **D. Ayyappan**, A. Gorny, W. Ye, E. Lobaton. "Machine Learning Techniques for Nematode Microscopic Image Analysis: A Systematic Review." *AgriEngineering*, 7(11):356, 2025. doi:10.3390/agriengineering7110356

[P4] P. K. Sharma, **D. Ayyappan**, G. Lazo, V. Tiwari. "Towards a Comprehensive Wheat Pangenome with Web-Based Visualization and Introductory AI." *Plant and Animal Genome Conference (PAG 31)*, 2024.

SELECTED PROJECTS

Automated Sim-to-Cloud Pipeline for 3D Assets | *SAM3D, Isaac Sim, AWS SageMaker* [GitHub]

- Reconstructed 3D meshes from single-view images (SAM3D), rendered synthetic datasets in **Isaac Sim** (Replicator API), and ran YOLOv8 inference on **AWS SageMaker** with S3 storage; automated deployment and data sync via GitHub Actions.

3D Brain Tumor Segmentation | *UNETR (ViT + U-Net), MONAI, PyTorch, Open3D* [GitHub]

- Implemented volumetric tumor segmentation with **UNETR** to improve tumor-class localization; rendered predictions in 3D for visualization.

TECHNICAL SKILLS

Languages: Python, C++, CUDA, MATLAB
3D Vision & Rendering: Gaussian Splatting, NeRF / Nerfstudio, COLMAP, Open3D, PyTorch3D, OpenCV, Blender, Unreal Engine
Simulation & Robotics: NVIDIA Warp (MPM), Newton, Isaac Sim, Genesis, Cosmos, ROS / ROS2, Raspberry Pi
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, metric learning, LLMs / LangChain
Tools: Git, Docker, AWS (SageMaker, S3), Linux, Jira, NumPy, Pandas, Matplotlib